Speech to Speech Translation Systems

Benchmarking and Implementation

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Introduction

A usual S2S (speech-to-speech) pipeline consists of 3 main components - ASR (Automatic Speech Recognition), MT (Machine Translation) and TTS (Text-to-Speech). The ASR module converts the input speech into text, the MT module translates the text from source to target language and the TTS module converts the translated text back into speech.

Previously, each stage was implemented separately and trained over different datasets. However, this leads to error propagation and information loss across the stages. Current state-of-the-art systems implement the complete pipeline as a unified system. This allows for better retention of prosodic and acoustic features across the stages, leading to more natural and accurate translations.

ASR: Overview

The ASR module of a S2S pipeline first converts the input speech into text. It involves several key components:

- **Audio Preprocessing:** Noise reduction and feature extraction using MFCCs, Mel spectrograms etc.
- **Encoder:** Responsible for capturing temporal dependencies and contextual information of the input audio using the extracted features. Typically implemented using RNNs, CNNs, or Transformers.
- **Decoder:** Generates the output text sequence based on the encoder reprensentation. Current state-of-the-art models often use attention mechanisms to focus on relevant parts of the input during decoding.

ASR Benchmarking: Setup

We run our benchmarking on the LibriSpeech dataset from which 100 samples of english speech (around 8 minutes of audio). The audio signals are resampled to 16 kHz and normalised to be all lowercase and without punctuation. The following models are evaluated:

- **OpenAI Whisper - Tiny:** Around 39M parameters
- **OpenAI Whisper - Base:** Around 74M parameters
- **OpenAI Whisper - Small:** Around 244M parameters
- **Distill Whisper - Small:** Around 166M parameters
- **OpenAI Whisper - Medium:** Around 394M parameters

We use word error rate (WER), character error rate (CER), sentence error rate (SER) and real-time factor (RTF) as our evaluation metrics.

ASR Benchmarking: WER Results

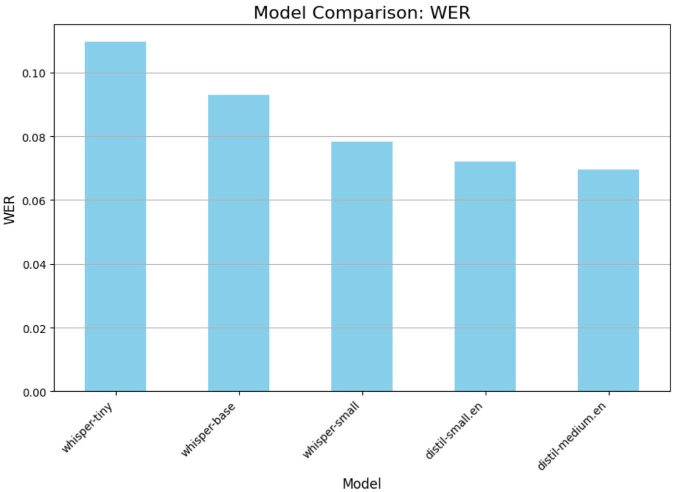


Figure: WER results for different ASR models

ASR Benchmarking: CER Results

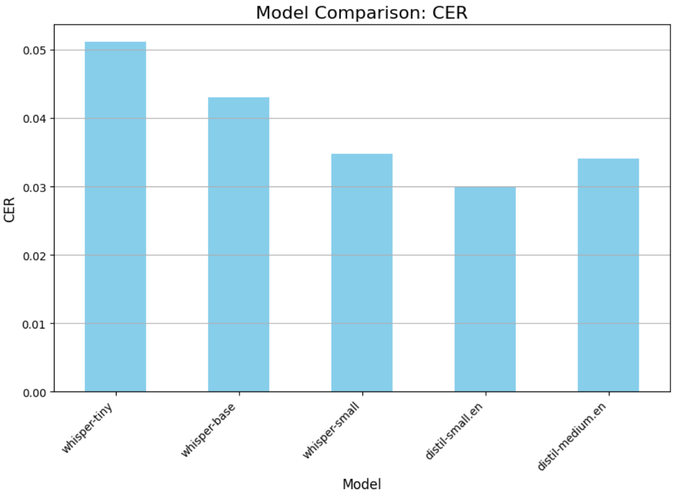


Figure: CER results for different ASR models

ASR Benchmarking: SER Results

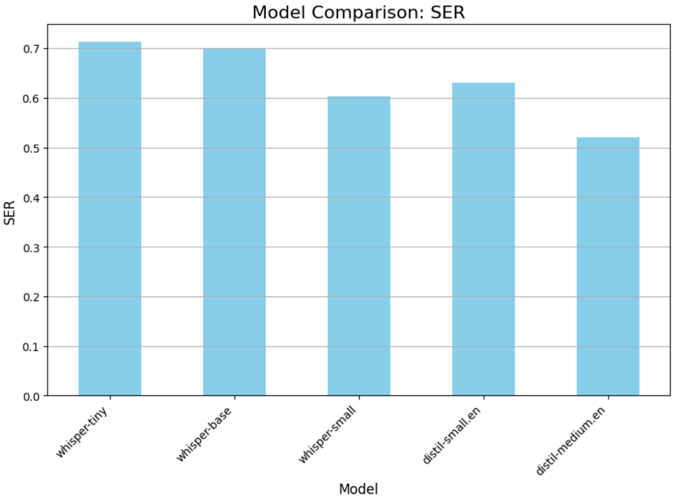


Figure: SER results for different ASR models

ASR Benchmarking: RTF Results

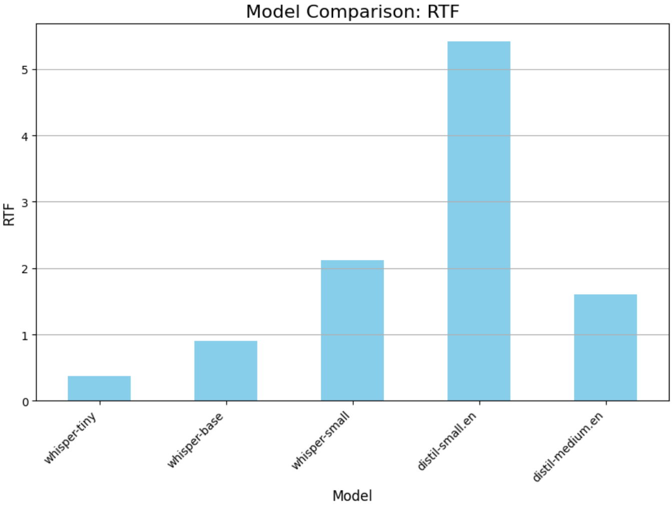


Figure: RTF results for different ASR models

ASR Benchmarking: Inferences

- Clear improvement in WER with increasing model size, with the medium model achieving the lowest WER of close to 0.07. The Distill small model performs better than OpenAI base model despite having fewer parameters.
- CER shows similar trends however the Distill medium model performs slightly worse.
- All models perform similary in terms of SER, with the Distill medium model performing the best.
- Distill small has the worst RTF. This is an anomalous result as the Distill medium performs much better with the same running conditions.

What is Machine Translation?

Definition

Machine Translation (MT) is the automated process of translating text from one natural language to another using computational models, without human intervention.

Key Paradigms:

- **Rule-Based MT (RBMT)** — hand-crafted linguistic rules
- **Statistical MT (SMT)** — phrase/word alignment probabilities
- **Neural MT (NMT)** — encoder–decoder deep learning
- **Large Language Models** — zero/few-shot translation

The English–Hindi Translation Challenge

Why English → Hindi?

- Hindi has **600M+ speakers** globally
- Significant demand in news, government, e-commerce
- Morphologically rich language with complex agreement
- **Low-resource** relative to European language pairs

Key Linguistic Challenges

- Script difference: Latin → Devanagari
- Free word order (SOV vs. SVO)
- Code-mixing in source text
- Cultural / idiomatic expressions

Test Set: IIT Bombay English–Hindi Corpus

A widely used benchmark dataset containing aligned sentence pairs sourced from diverse domains including news, government documents, and general text.

Evaluation Metrics Overview

Lexical / N-gram Metrics

- **BLEU** (Bilingual Evaluation Understudy) — precision of *n*-grams vs. reference; ↑ higher is better
- **chrF++** — character *n*-gram F-score; more robust for morphology; ↑
- **TER** (Translation Edit Rate) — edit operations to reach reference; ↓ lower is better
- **METEOR** — alignment-based with synonymy; ↑

Neural / Semantic Metric

- **BERTScore-F1** — contextual embeddings from BERT; captures semantic similarity beyond surface form; ↑

Metric Limitations

No single metric perfectly captures translation quality. Human evaluation remains the gold standard, but automated metrics enable fast, reproducible comparison at scale.

Systems Under Evaluation

MarianMT (Helsinki-NLP)

- Encoder–decoder Transformer
- Lightweight, fast inference
- Widely used baseline in research

mBART-50 (Meta AI)

- Multilingual BART pre-training
- Fine-tuned on 50 languages
- Denoising pre-training objective

NLLB-600M (Meta AI)

- 200-language multilingual model
- 600M parameter variant
- Dedicated low-resource language support

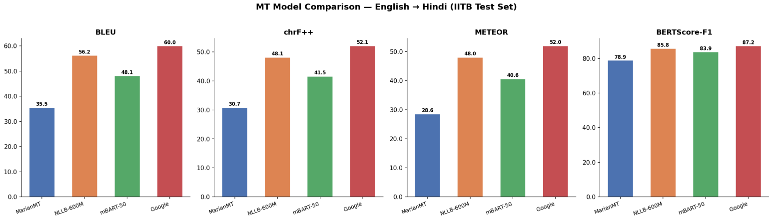
Google Translate (API)

- Commercial neural MT system
- Massive proprietary training data
- Industry gold-standard reference

Overall Metric Scores

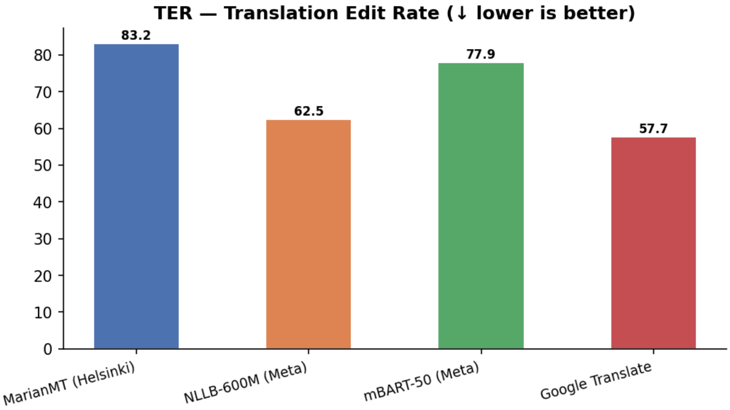
Model	BLEU	chrF++	TER	METEOR	BERT-F1
MarianMT	35.50	30.72	83.15	28.56	78.92
mBART-50	48.12	41.53	77.91	40.60	83.86
NLLB-600M	56.20	48.11	62.52	48.02	85.85
Translate	**60.00**	**52.07**	**57.70**	**51.97**	**87.25**

Metrics Comparison Chart



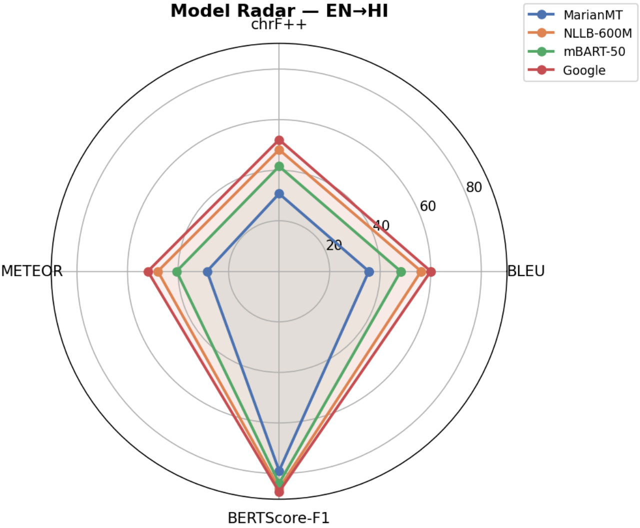
Bar chart of all five metrics across the four systems. Google Translate leads on all metrics; MarianMT significantly lags. NLLB-600M is the strongest open-source system.

TER Comparison



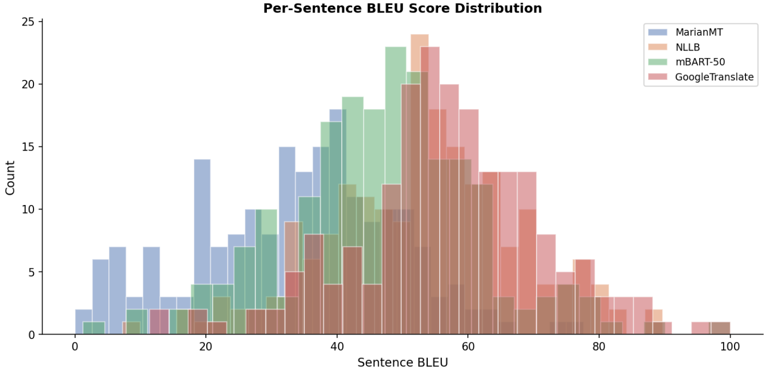
Translation Edit Rate (TER) — **lower is better**. *Google Translate requires fewest edits (57.70) to match the reference. MarianMT requires the most edits (83.15), indicating lower fluency.*

Radar Chart — Multi-Metric Profile



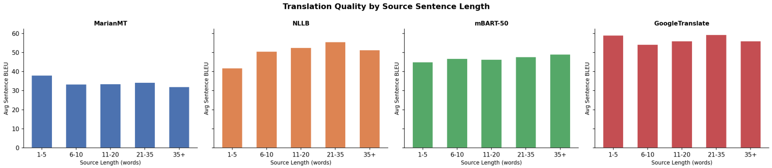
Radar chart showing relative normalised performance across all five metrics. Google Translate occupies the outermost position consistently; NLLB-600M is the

BLEU Score Distribution



Per-sentence BLEU score distributions. All models show right-skewed distributions with a long tail of low-scoring sentences, reflecting the inherent difficulty of certain sentence types.

Sentence Length vs. BLEU Score



BLEU score as a function of source sentence length. Short sentences (1–5 tokens) generally yield higher BLEU, while longer sentences (21+ tokens) trend lower due to compounding translation errors.

Hardest & Easiest Sentences

5 Hardest Sentences (avg BLEU)

- *"The water rail was fit and well by the time it was released."* — Idiomatic, rare vocabulary
- *"The rep adds: 'Perhaps she is confusing him...' "* — Quotation within quotation
- *"Kaplowitz contends the premise that holds the most weight is the epidemic of obesity."* — Complex nominal structure
- *"They never believed he died the way the sheriff concluded."* — Relative clause, named entity
- *"Court blocks ruling on NYPD stop-and-frisk policy"* — Headline, named entities

5 Easiest Sentences (avg BLEU)

- *"He did not tell anyone that Jahid has drowned."* — Simple, high-frequency vocabulary
- *"Today he will meet the families of the people who died in Patna last Sunday."* — Direct news style
- *"A southern Georgia judge on Wednesday ordered authorities to release all surveillance video."* — Factual
- *"He said: 'I really want to develop an invisible ice cream.' "* — Simple quoted speech
- *"They asked some subjects to count the number of dots on a computer screen."* — Simple active

Conclusions

Main Findings

1. **Google Translate** achieves the best performance across all five metrics (BLEU 60.0, BERTScore 87.25)
2. **NLLB-600M** is the best open-source alternative (BLEU 56.2) and viable for production use cases
3. **mBART-50** occupies a middle ground — acceptable but beaten clearly by NLLB
4. **MarianMT** provides a useful baseline but is not competitive for high-quality EN→HI translation

Recommendations

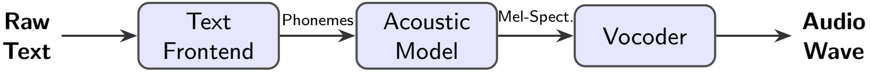
- Use **Google Translate API** when accuracy is critical
- Use **NLLB-600M** for offline, privacy-sensitive, or cost-sensitive deployments
- Consider **domain fine-tuning** of NLLB for specialised corpora (medical, legal)

What is Text-to-Speech (TTS)?

Text-to-Speech (TTS) is the artificial synthesis of human-sounding speech from raw text.

- **Core Objectives:**
  - *Intelligibility:* Clear, understandable pronunciation of words.
  - *Naturalness:* Accurate prosody, intonation, rhythm, and emotion.
- **Evolution of the Technology:**
  - **Concatenative (1990s-2000s):** Splicing together tiny fragments of pre-recorded human speech. Highly intelligible but sounded robotic and rigid.
  - **Parametric (2000s-2010s):** Statistical models (like HMMs) generating audio based on parameters. Smoother, but sounded muffled or "buzzy".
  - **Neural (Present):** Deep learning models predicting high-fidelity audio waves directly from text, achieving near-human naturalness.

The Standard Neural TTS Pipeline



- **Text Frontend:** Cleans and normalizes text (e.g., expanding "180" to "one hundred eighty") and performs Grapheme-to-Phoneme (G2P) conversion to determine pronunciation.
- **Acoustic Model:** Maps phonemes to acoustic representations, predicting pitch, energy, and duration. *(Examples: Tacotron 2, FastSpeech)*
- **Vocoder:** Reconstructs the actual time-domain audio waveform from the highly compressed Mel-spectrogram. *(Examples: WaveNet, HiFi-GAN)*

Systems Under Evaluation

MMS-TTS

- **Architecture:** VITS-based (End-to-End).
- **Primary Strength:** Highly lightweight and efficient; state-of-the-art for low-resource languages.
- **Limitation:** Generally single-speaker per model with fixed prosody.

Parler-TTS

- **Architecture:** Autoregressive Transformer.
- **Primary Strength:** Granular control over emotion, tone, pacing, and background noise via text descriptions.
- **Limitation:** Higher inference latency due to autoregressive generation.

XTTS-v2

- **Architecture:** GPT-style + Mel/HiFi-GAN.
- **Primary Strength:** Requires only $\sim$3 seconds of reference audio; supports cross-lingual voice cloning.
- **Limitation:** High resource consumption compared to non-autoregressive models.

Evaluation Metrics for TTS Models

Intelligibility & Efficiency

- **Word Error Rate (WER)**
  What it is: Transcription accuracy of the audio using an ASR system. Lower is better.
- **Character Error Rate (CER)**
  What it is: Similar to WER, but captures finer phonetic/spelling errors. Lower is better.
- **Real-Time Factor (RTF)**
  What it is: Processing time divided by generated audio length. Lower is better.

Prosody & Expressiveness

- **Pitch Variance (F_0 Std Dev)**
  What it is: Range of intonation.
- **Energy Dynamics (RMS Std)**
  What it is: Variations in volume/loudness.
- **Speaking Rate**
  What it is: Speed (Syllables/words per sec).
- **Pause Ratio (Onset/s)**
  What it is: Frequency of breath/silence pauses.

TTS: Benchmarking Results

Model	Language	Latency (ms)	RTF	Throughput (cps)
MMS-TTS	English	2621.382	0.439	31.711
MMS-TTS	Hindi	2451.2891	0.432	27.49
Parler-TTS	English	68026.311	11.264	1.281
XTTS-v2	English	26182.9	3.438	3.491
XTTS-v2	Hindi	23687.88	3.204	2.718

Table: Benchmarking results for TTS models across Performance Metrics.

TTS: Benchmarking Results

Model	Language	WER	CER
MMS-TTS	English	34.892	17.968
MMS-TTS	Hindi	-	-
Parler-TTS	English	27.57	19.854
XTTS-v2	English	20.465	6.825
XTTS-v2	Hindi	-	-

Table: Benchmarking results for TTS models across Performance Metrics (Contd.).

**WER/CER for Hindi were turning out to be extremely high due to ASR model's poor performance on Hindi audio, so we have omitted those values to avoid skewing the analysis.*

TTS: Benchmarking Results

Model	Language	Pitch (Mean)	Pitch (Std)	Pitch Range
MMS-TTS	English	99.978	19.158	81.388
MMS-TTS	Hindi	148.216	37.359	176.392
Parler-TTS	English	233.055	54.52	253.809
XTTS-v2	English	141.029	29.674	136.078
XTTS-v2	Hindi	139.625	16.915	76.358

Table: Benchmarking results for TTS models across Prosodic Metrics (Contd.).

TTS: Benchmarking Results

Model	Language	Speaking Rate	Energy Dynamics	Pause Ratio
MMS-TTS	English	6.788	0.067	0.183
MMS-TTS	Hindi	5.808	0.093	0.246
Parler-TTS	English	5.35	0.082	0.153
XTTS-v2	English	4.284	0.099	0.212
XTTS-v2	Hindi	4.268	0.1002	0.186

Table: Benchmarking results for TTS models across Prosodic Metrics.

TTS: Plots

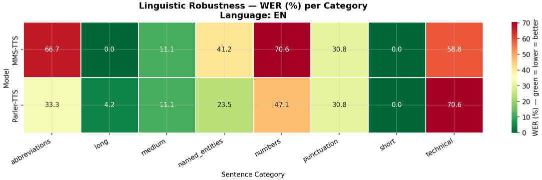


Figure: WER Heatmap for English for MMS-TTS and Parler-TTS for different types of sentences.

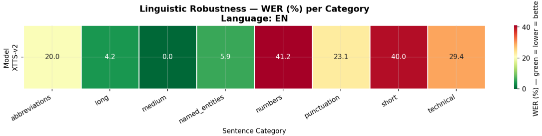


Figure: WER Heatmap for English for XTTS-v2 for different types of sentences.

End-to-End Pipeline

We also implemented a full S2S pipeline using few of the models that we benchmarked here and compared them against the existing SeamlessM4T-V2 (2.3B parameters) model. The models that we used for the pipeline were:

- **ASR:** OpenAI Whisper - Base (74M parameters)
- **MT:** Helsinki-NLP Opus-MT (English to Hindi) (77M parameters)
- **TTS:** MMS-TTS by Meta AI (559M parameters)

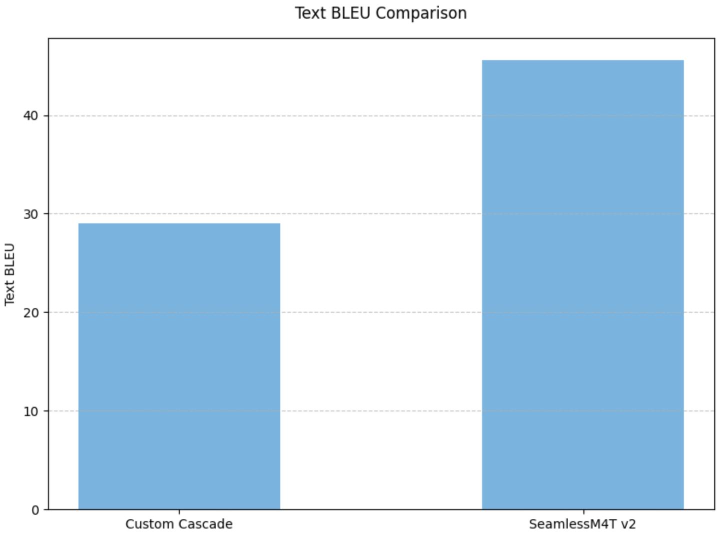
We use 8 predefined sentences as input to the pipeline and evaluate the output on various metrics. All audios were sampled at 16khz and dynamically aligned to ensure fair comparison between input and output audio for acoustic metrics.

S2S Pipeline: Scoring Metrics

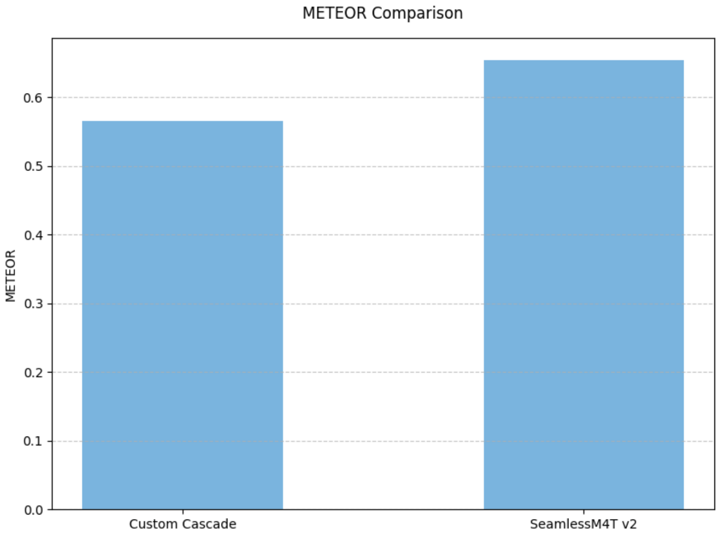
We use the following scoring metrics for the end-to-end pipeline:

- **METEOR:** Measures n-gram precision and recall while accounting for synonyms and stemming.
- **chrF++:** Extends character n-gram F-score with word-level n-grams, capturing both fine-grained and coarse-grained translation quality.
- **RTF:** Real-time factor, measuring the speed of the system relative to real-time processing.
- **Latency:** Time taken for the system to process and respond to input.
- **Text BLEU:** Measures the quality of the transcribed text from ASR against the original input text.
- **F0 Pitch Correlation:** Measures the similarity between the pitch between in input and output audio.
- **Energy RMS Correlation:** Measures the similarity between the loudness between in input and output audio.
- **Duration Ratio:** Ratio between the output and input audio length.

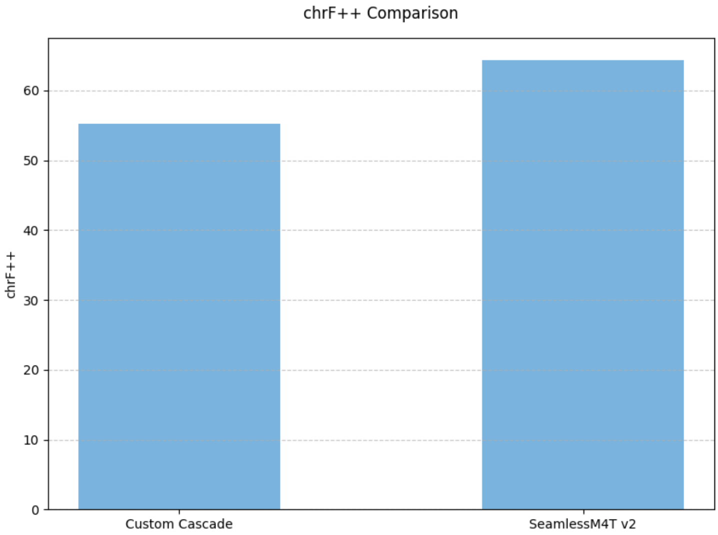
S2S Pipeline: Results - Text BLEU



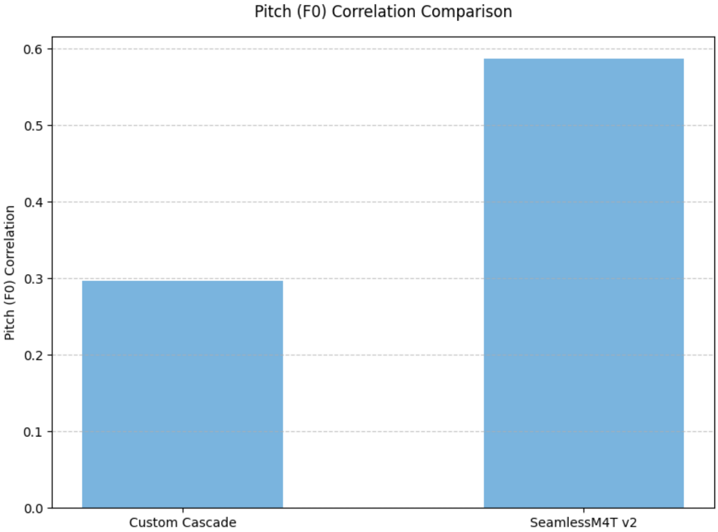
S2S Pipeline: Results - METEOR



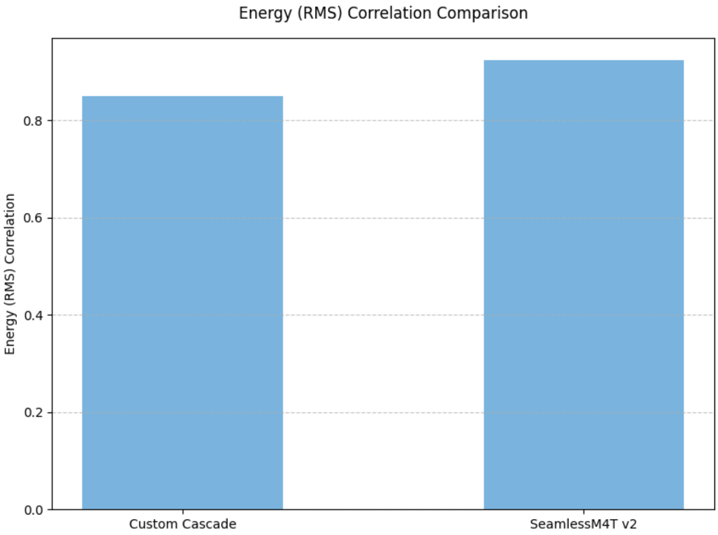
S2S Pipeline: Results - chrF++



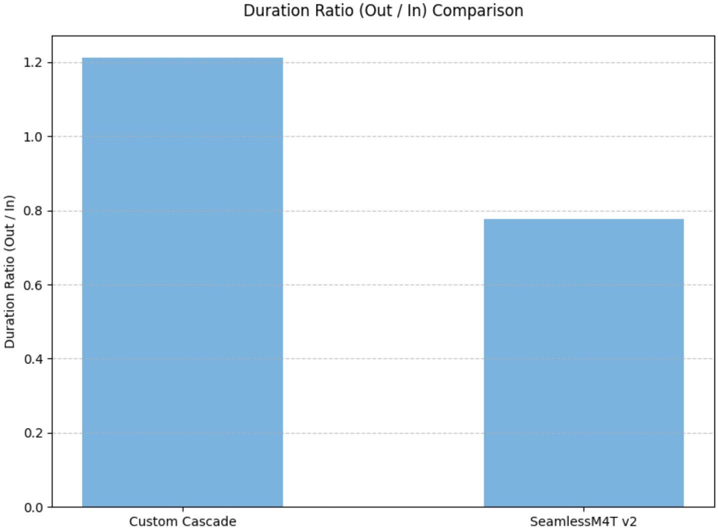
S2S Pipeline: Results - F0 Pitch Correlation



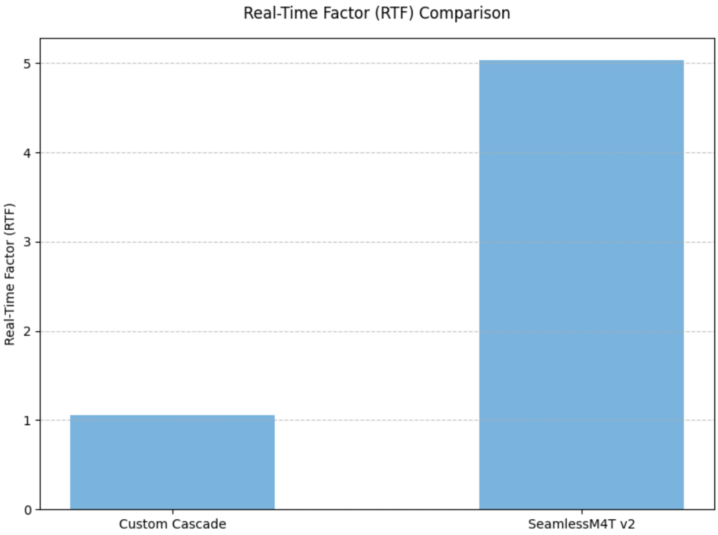
S2S Pipeline: Results - Energy RMS Correlation



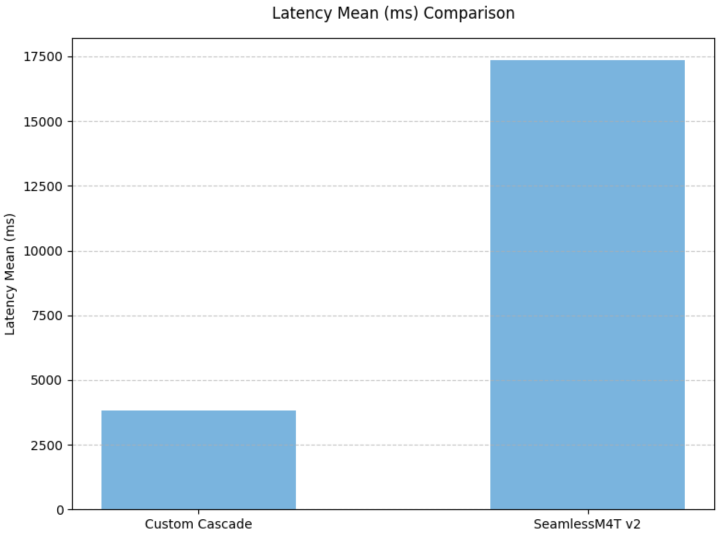
S2S Pipeline: Results - Duration Ratio



S2S Pipeline: Results - RTF



S2S Pipeline: Results - Latency



S2S Pipeline: Conclusions

- SeamlessM4T- V2 outperforms the custom pipeline over the metrics by still has a lot of room for improvement in terms of latency and RTF.
- The input audio is that of a female speaker which is matched well by the SeamlessM4T-V2 model but not as well by the custom pipeline which outputs the audio in a male voice.
- We see that if an error is made in the ASR stage, it propagates through the pipeline and results in a much lower METEOR and chrF++ score.

Thank You

Thank you!